Notes: Chow Theory for Matroid Deletion

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1. Algebraic Preliminaries

1.1. The Module of Flats and Perverse Subgroups

Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. For any integer n, we denote by $\mathrm{Pal}(n)$ the additive group of polynomials $f(x) \in R$ satisfying $f(x) = x^n f(x^{-1})$. While $\mathrm{Pal}(0)$ is a subring of R, we note that $\mathrm{Pal}(n)$ for $n \neq 0$ is not, as $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subseteq \mathrm{Pal}(n+m)$.

Let M be a matroid with lattice of flats $\mathcal{L}(\mathrm{M})$.

Definition 1. *The module $\mathcal{H}(\mathrm{M})$ is the free R-module with a standard basis $\{\mathrm{F}\}_{F \in \mathcal{L}(\mathrm{M})}$. An element $\alpha \in \mathcal{H}(\mathrm{M})$ is a formal sum*

$$\alpha = \sum_{F \in \mathcal{L}(\mathrm{M})} \alpha_F \cdot \mathrm{F}, \quad \text{where } \alpha_F \in R.$$

We also define the R-linear maps $S_F : \mathcal{H}(\mathrm{M}) \to R$ for any flat $F \in \mathcal{L}(\mathrm{M})$ by

$$S_F(\alpha) := \sum_{G \geq F} x^{\mathrm{rk}(F) - \mathrm{rk}(G)} \alpha_G.$$

Following Braden-Vysogorets, there is an important abelian subgroup of $\mathcal{H}(\mathrm{M})$.

Definition 2. *The subgroup of **perverse elements** $\mathcal{H}_p(\mathrm{M}) \subset \mathcal{H}(\mathrm{M})$ consists of all α such that for every flat $F \in \mathcal{L}(\mathrm{M})$:*

1. *The coefficient α_F is a polynomial, i.e., $\alpha_F \in \mathbb{Z}[x]$.*
2. *The Laurent polynomial $S_F(\alpha)$ is in $\mathrm{Pal}(0)$.*

Definition 3. *We relax the first condition to define a larger subgroup $\mathcal{H}'_p(\mathrm{M}) \subseteq \mathcal{H}(\mathrm{M})$ consisting of all elements α that satisfy $S_F(\alpha) \in \mathrm{Pal}(0)$ for all $F \in \mathcal{L}(\mathrm{M})$.*

1.2. The Chow and Kazhdan-Lusztig Bases

The module $\mathcal{H}(\mathrm{M})$ has natural bases constructed from the Kazhdan-Lusztig and Chow polynomials.

Definition 4. *The **Kazhdan-Lusztig basis** $\{\zeta^F\}$ is defined by*

$$\zeta^F = \sum_{G \leq F} \zeta^F_G \cdot \mathrm{G} := \sum_{G \leq F} x^{\mathrm{rk}(F) - \mathrm{rk}(G)} \mathrm{P}_{\mathrm{M}^F_G}(x^{-2}) \cdot \mathrm{G},$$

where $\mathrm{P_M}$ *is the Kazhdan-Lusztig polynomial of* M.

Remark. Braden-Vysogorets prove that the elements $\{\zeta^F\}$ form a $\mathbb{Z}$-basis for $\mathcal{H}_p(\mathrm{M})$. In particular, if $\alpha = \sum_F \alpha_F \cdot \mathrm{F} \in \mathcal{H}_p(\mathrm{M})$, then

$$\alpha = \sum_F \alpha_F(0) \cdot \zeta^F.$$

Definition 5. *The* ***Chow basis*** $\{\mathrm{c}^F\}$ *is defined by*

$$\mathrm{c}^F = \sum_{G \leq F} \mathrm{c}^F_G \cdot \mathrm{G} := \sum_{G \leq F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)} \underline{\mathrm{H}}_{\mathrm{M}^F_G}(x^{-2}) \cdot \mathrm{G},$$

where $\underline{\mathrm{H}}_{\mathrm{M}}$ *is the Chow polynomial of* M*, which is palindromic of degree* $\mathrm{rk}(\mathrm{M}) - 1$.

Remark. The coefficients of the Chow basis have a special structure:

$$\mathrm{c}^F_F = 1 \in \mathrm{Pal}(0), \quad \text{and} \quad \mathrm{c}^F_G \in \mathrm{Pal}(2) \text{ for all } G < F.$$

Definition 6. *We define the module* $\mathcal{S}(\mathrm{M})$ *to be the free* $\mathrm{Pal}(0)$*-module with basis* $\{\mathrm{c}^F\}$.

Definition 7. *For an element* e *in the ground set of* M*, the* ***deletion operator*** *is an* R*-module homomorphism* $\Delta : \mathcal{H}(\mathrm{M}) \to \mathcal{H}(\mathrm{M} \setminus e)$ *defined by*

$$\Delta(\mathrm{F}) = x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e}, \quad \textit{where } \delta(F) := \mathrm{rk}_{\mathrm{M}}(F) - \mathrm{rk}_{\mathrm{M}\setminus e}(F \setminus e).$$

Remark. The deletion operators commute: $\Delta_e \circ \Delta_f = \Delta_f \circ \Delta_e : \mathcal{H}(\mathrm{M}) \to \mathcal{H}(\mathrm{M} \setminus \{e, f\})$.

Definition 8. *For any element* e *in the ground set of* M*, we define the set*

$$S_e(\mathrm{M}) := \{F \in \mathcal{L}(\mathrm{M}) \mid e \notin F \text{ and } F \cup e \in \mathcal{L}(\mathrm{M})\}.$$

As demonstrated in Braden-Vysogorets, $S_e(\mathrm{M})$ *plays a crucial role in the deletion formula.*

2. Main Propositions

Proposition 9. *For all* $F \in \mathcal{L}(\mathrm{M})$, $\mathrm{c}^F \in \mathcal{H}'_p(\mathrm{M})$*, i.e., it satisfies* $S_G(\mathrm{c}^F) \in \mathrm{Pal}(0)$ *for all* $G \in \mathcal{L}(\mathrm{M})$.

Proof. We see that, for any $G \in \mathcal{L}(\mathrm{M})$,

$$\begin{aligned} S_G(\mathrm{c}^F) = \sum_{H : G \leq H \leq F} x^{\mathrm{rk}(G)-\mathrm{rk}(H)} \cdot \mathrm{c}^F_H &= \sum_{H : G \leq H \leq F} x^{\mathrm{rk}(G)-\mathrm{rk}(H)} \left(x^{\mathrm{rk}(F)-\mathrm{rk}(H)} \underline{\mathrm{H}}_{\mathrm{M}^F_H}(x^{-2}) \right) \\ &= x^{\mathrm{rk}(F)-\mathrm{rk}(G)} \sum_{H : G \leq H \leq F} x^{-2(\mathrm{rk}(H)-\mathrm{rk}(G))} \underline{\mathrm{H}}_{\mathrm{M}^F_H}(x^{-2}). \end{aligned}$$

This sum is precisely the definition of the *augmented* Chow polynomial, $\mathrm{H}_{\mathrm{M}^F_G}(x^{-2})$. So, $S_G(c^F) = x^{\mathrm{rk}(F)-\mathrm{rk}(G)} \mathrm{H}_{\mathrm{M}^F_G}(x^{-2})$. Let $d = \mathrm{rk}(F) - \mathrm{rk}(G)$. The term x^d is in $\mathrm{Pal}(2d)$. The term $\mathrm{H}_{\mathrm{M}^F_G}(x^{-2})$ is in $\mathrm{Pal}(-2d)$. The product is in $\mathrm{Pal}(2d - 2d) = \mathrm{Pal}(0)$ as required. □

Proposition 10. *The subgroup* $\mathcal{H}'_p(\mathrm{M})$ *is a* $\mathrm{Pal}(0)$*-module.*

Proof. Let $\alpha = \sum_F p_F \cdot \mathrm{F} \in \mathcal{H}'_p(\mathrm{M})$ and $q \in \mathrm{Pal}(0)$.

We see that

$$S_G(q \cdot \alpha) = \sum_{H \geq G} x^{\mathrm{rk}(G)-\mathrm{rk}(H)}(q \cdot p_H) = q \cdot S_G(\alpha) \in \mathrm{Pal}(0).$$

Therefore, $q \cdot \alpha \in \mathcal{H}'_p(\mathrm{M})$. □

Proposition 11. *Any Laurent polynomial f can be uniquely written as a sum $f = f_0 + f_2$ where $f_0 \in \mathrm{Pal}(0)$ and $f_2 \in \mathrm{Pal}(2)$. The projections are given by*

$$f_0(x) = \frac{x^2 f(x^{-1}) - f(x)}{x^2 - 1} \quad \textit{and} \quad f_2(x) = \frac{f(x) - f(x^{-1})}{1 - x^{-2}}.$$

Proposition 12. $\mathcal{S}(\mathrm{M}) = \mathcal{H}'_p(\mathrm{M})$ *as* $\mathrm{Pal}(0)$*-modules.*

Proof. It is enough to show that $\mathcal{H}'_p(\mathrm{M}) \subseteq \mathcal{S}(\mathrm{M})$.

Let $\alpha = \sum_F \alpha_F \cdot \mathrm{F}$ be an arbitrary element of $\mathcal{H}'_p(\mathrm{M})$. Since $\{\mathrm{c}^F\}$ is a basis for the entire module $\mathcal{H}(\mathrm{M})$, α has a unique expansion in this basis:

$$\alpha = \sum_{F \in \mathcal{L}(\mathrm{M})} q_F \cdot \mathrm{c}^F \quad \text{for } q_F \in \mathbb{Z}[x, x^{-1}].$$

To show that $\alpha \in \mathcal{S}(\mathrm{M})$, we must prove that each of these coefficients q_F is actually in $\mathrm{Pal}(0)$. We prove this by induction on the corank of F.

Base Case: The relationship between the coefficients is given by the change-of-basis formula:

$$\alpha_F = \sum_{G \geq F} q_G \cdot \mathrm{c}^G_F = q_F + \sum_{G > F} q_G \cdot \mathrm{c}^G_F.$$

For $F = E$, this simplifies to $q_E = \alpha_E$. Since $\alpha \in \mathcal{H}'_p(\mathrm{M})$, we have $S_E(\alpha) = \alpha_E \in \mathrm{Pal}(0)$. This implies that $q_E \in \mathrm{Pal}(0)$, establishing the base case.

Inductive Step: Assume that for all flats G with $\mathrm{rk}(G) > \mathrm{rk}(F)$, the coefficient q_G is in $\mathrm{Pal}(0)$.

Since $\alpha \in \mathcal{H}'_p(\mathrm{M})$, it satisfies:

$$S_F(\alpha) = \sum_{H \geq F} x^{\mathrm{rk}(F)-\mathrm{rk}(H)} \alpha_H \in \mathrm{Pal}(0).$$

We substitute the change-of-basis formula $\alpha_H = \sum_{L \geq H} q_L \cdot \mathrm{c}^L_H$ into this expression:

$$S_F(\alpha) = \sum_{H \geq F} x^{\mathrm{rk}(F)-\mathrm{rk}(H)} \left(\sum_{L \geq H} q_L \cdot \mathrm{c}^L_H \right) = \sum_{L \geq F} q_L \left(\sum_{H : F \leq H \leq L} x^{\mathrm{rk}(F)-\mathrm{rk}(H)} \mathrm{c}^L_H \right) = \sum_{L \geq F} q_L \cdot S_F(\mathrm{c}^L).$$

We isolate the $L = F$ term and rearrange the sum:

$$q_F = S_F(\alpha) - \sum_{L > F} q_L \cdot S_F(\mathrm{c}^L).$$

By the inductive hypothesis, each q_L in the sum is in $\mathrm{Pal}(0)$. Since $\mathrm{Pal}(0)$ is a ring, the difference on the right is in $\mathrm{Pal}(0)$. Thus, q_F must also be in $\mathrm{Pal}(0)$. □

Proposition 13 (Change of Basis Formula). *If $\alpha = \sum_F \alpha_F \cdot \mathrm{F} \in \mathcal{S}(\mathrm{M})$, then its expansion in the c^F basis is $\alpha = \sum_F \overline{\alpha_F} \cdot \mathrm{c}^F$, where $\overline{\alpha_F}$ is the unique $\mathrm{Pal}(0)$ component of α_F from the decomposition in Proposition 11.*

Proof. Let $\alpha \in \mathcal{S}(\mathrm{M})$. By the definition of $\mathcal{S}(\mathrm{M})$, α can be uniquely expressed as a $\mathrm{Pal}(0)$-linear combination of the c^F basis vectors:

$$\alpha = \sum_{F \in \mathcal{L}(\mathrm{M})} q_F \cdot \mathrm{c}^F, \quad \text{where each } q_F \in \mathrm{Pal}(0).$$

We can also express α in the standard basis as $\alpha = \sum_F \alpha_F \cdot \mathrm{F}$. As in the proof of proposition 12, the change-of-basis formula relating the two sets of coefficients is:

$$q_F = \alpha_F - \sum_{G > F} q_G \cdot \mathrm{c}^G_F. \tag{1}$$

The goal is to show that $q_F = \overline{\alpha_F}$. We prove this by induction on the corank of F as in the previous proposition.

Base Case: For the base case, we see that $q_E = \alpha_E = S_E(\alpha) \in \mathrm{Pal}(0)$. This implies that $q_E = \overline{\alpha_E}$.

Inductive Step: Assume that for all flats G with $\mathrm{rk}(G) > \mathrm{rk}(F)$, the coefficient $q_G = \overline{\alpha_G}$.

Then the entire sum $\sum_{G>F} q_G \cdot \mathrm{c}^G_F$ is in $\mathrm{Pal}(2)$, since for any $G > F$, the coefficient q_G is in $\mathrm{Pal}(0)$ and the basis coefficient c^G_F is in $\mathrm{Pal}(2)$, as noted in a previous remark. This shows that α_F is the sum of q_F (which, by definition, is in $\mathrm{Pal}(0)$) and another term that is in $\mathrm{Pal}(2)$.

By Proposition 11, this decomposition is unique. Therefore, the coefficient q_F must be the unique $\mathrm{Pal}(0)$ component of the α_F. □

Proposition 14. *For any flat $F \in \mathcal{L}(\mathrm{M})$, the element $\Delta(\mathrm{c}^F)$ is in $\mathcal{S}(\mathrm{M} \setminus e)$.*

Proof. Let $F \in \mathcal{L}(\mathrm{M})$ be an arbitrary flat. We need to show that $S_H^{\mathrm{M}\setminus e}(\Delta(\mathrm{c}^F))$ is in $\mathrm{Pal}(0)$ for all $H \in \mathcal{L}(\mathrm{M} \setminus e)$.

Let $\beta = \Delta(\mathrm{c}^F) = \sum_{G \in \mathcal{L}(\mathrm{M}\setminus e)} \beta_G \cdot \mathrm{G}$.

By definition, $\mathrm{c}^F = \sum_{L \le F} \mathrm{c}^F_L \cdot \mathrm{L}$. Applying the Δ operator gives:

$$\beta = \Delta(\mathrm{c}^F) = \sum_{L \le F} \mathrm{c}^F_L x^{-\delta(L)} \cdot \mathrm{L} \setminus \mathrm{e}.$$

The coefficient β_G is the sum over all flats $L \le F$ such that $L \setminus e = G$. Thus, we have:

$$\begin{aligned} S_H^{\mathrm{M}\setminus e}(\Delta(\mathrm{c}^F)) = \sum_{G \ge H} x^{\mathrm{rk}(H)-\mathrm{rk}(G)} \beta_G &= \sum_{G \ge H} x^{\mathrm{rk}(H)-\mathrm{rk}(G)} \left(\sum_{L : L \le F, L\setminus e = G} \mathrm{c}^F_L x^{-\delta(L)} \right) \\ &= \sum_{L : L \le F, L\setminus e \ge H} x^{\mathrm{rk}(H)-\mathrm{rk}(L\setminus e)} \mathrm{c}^F_L x^{-\delta(L)} \end{aligned}$$

The exponent on x simplifies to $\mathrm{rk}(H) - \mathrm{rk}(L)$ using the definition of discrepancy.

$$S_H^{\mathrm{M}\setminus e}(\Delta(\mathrm{c}^F)) = \sum_{L : L \le F, L\setminus e \ge H} x^{\mathrm{rk}(H)-\mathrm{rk}(L)} \mathrm{c}^F_L.$$

Now we apply Lemma 2.15 from Braden-Vysogorets. The condition $L \setminus e \geq H$ is equivalent to $L \geq \overline{H}$, where $\overline{H}$ is the closure of H in M. Furthermore, $\mathrm{rk}_{\mathrm{M}\setminus e}(H) = \mathrm{rk}_{\mathrm{M}}(\overline{H})$.

Substituting these two facts, the summation range becomes $L \in [\overline{H}, F]$ and the expression is:

$$S_H^{\mathrm{M}\setminus e}(\Delta(\mathrm{c}^F)) = \sum_{L:\overline{H}\leq L\leq F} x^{\mathrm{rk}(\overline{H})-\mathrm{rk}(L)}\mathrm{c}_L^F = S_{\overline{H}}(\mathrm{c}^F) \in \mathrm{Pal}(0).$$

Since the palindromic condition holds for any flat H, we have proven that $\Delta(\mathrm{c}^F) \in \mathcal{S}(\mathrm{M}\setminus e)$. □

Remark. The map $\Delta : \mathcal{S}(\mathrm{M}) \to \mathcal{S}(\mathrm{M}\setminus e)$ is a $\mathrm{Pal}(0)$-module homomorphism that restricts from the $\mathbb{Z}[x, x^{-1}]$-module homomorphism $\Delta : \mathcal{H}(\mathrm{M}) \to \mathcal{H}(\mathrm{M}\setminus e)$.

Proposition 15. *For any flat $F \in \mathcal{L}(\mathrm{M})$, the following identities hold in $\mathcal{S}(\mathrm{M}\setminus e)$:*

1. *If e is not a coloop in M^F, then*

$$\Delta(\mathrm{c}^F) = \mathrm{c}^{F\setminus e} + x^{-1} \sum_{G\in S_e(\mathrm{M}^F)} \mathrm{c}^F_{G\cup e} \cdot \mathrm{c}^G \tag{2}$$

2. *If e is a coloop in M^F, then*

$$\Delta(\mathrm{c}^F) = (x + x^{-1})\mathrm{c}^{F\setminus e} + x^{-1} \sum_{G\in S_e(\mathrm{M}^F), G\neq F\setminus e} \mathrm{c}^F_{G\cup e} \cdot \mathrm{c}^G \tag{3}$$

Proof. By Proposition 13, the coefficients q_G in the expansion $\Delta(\mathrm{c}^F) = \sum q_G \cdot \mathrm{c}^G$ are the unique $\mathrm{Pal}(0)$ components of the standard-basis coefficients,

$$\beta_G = \sum_{L:L\leq F, L\setminus e=G} \mathrm{c}^F_L x^{-\delta(L)}.$$

Thus we analyze q_G by considering different cases for the flat G.

Coefficients for $G \neq F \setminus e$: For any flat G not equal to $F \setminus e$, its preimages $L \leq F$ are a subset of $\{\overline{G}, G \cup e\}$ where $\overline{G}$ is the closure of G in M.

- If $G \in S_e(\mathrm{M}^F)$, $\overline{G} = G$ and the standard-basis coefficient is $\beta_G = \mathrm{c}^F_G + x^{-1}\mathrm{c}^F_{G\cup e}$. This is a sum of a $\mathrm{Pal}(2)$ term and a $\mathrm{Pal}(0)$ term. Its unique $\mathrm{Pal}(0)$ part is $q_G = x^{-1}\mathrm{c}^F_{G\cup e}$.
- If $G \notin S_e(\mathrm{M}^F)$, the only possible preimage is $L = \overline{G}$. The standard-basis coefficient is $\beta_G = \mathrm{c}^F_{\overline{G}}$, which is in $\mathrm{Pal}(2)$. Its $\mathrm{Pal}(0)$ part is $q_G = 0$.

These coefficients account for the summation in both formulas of the proposition.

Coefficient for $G = F \setminus e$: Here we analyze $\beta_{F\setminus e}$ by considering the the following two cases:

- If e is **not a coloop** in M^F, the only flat $L \leq F$ that deletes to $F \setminus e$ is $L = F$. The standard-basis coefficient is $\beta_{F\setminus e} = \mathrm{c}^F_F x^{-0} = 1 \in \mathrm{Pal}(0)$. So $q_{F\setminus e} = 1$, proving the first part of the proposition.
- If e is **a coloop** in M^F, the preimages are $L = F \setminus e$ and $L = F$. The standard-basis coefficient is $\beta_{F\setminus e} = \mathrm{c}^F_{F\setminus e} + x^{-1}\mathrm{c}^F_F = x + x^{-1} \in \mathrm{Pal}(0)$. So $q_{F\setminus e} = x + x^{-1}$. This proves the second part of the proposition.

□

3. Recovery of the Semi-Small Decomposition Formulas

We now show how the formulas in Proposition 15 recover the known semi-small decomposition formulas for the Chow polynomial $\underline{\mathrm{H}}$ from [FMSV24]. This is achieved by equating the coefficients of the empty flat $\emptyset$ on both sides of the identities.

Let $f = \mathrm{rk}(F)$. The coefficient of $\emptyset$ in $\Delta(\mathrm{c}^F)$ is:

$$\beta_\emptyset = \mathrm{c}^F_\emptyset x^{-\delta(\emptyset)} + \mathrm{c}^F_{\{e\}} x^{-\delta(\{e\})} = x^f \underline{\mathrm{H}}_{\mathrm{M}^F}(x^{-2}) + x^{f-2}\underline{\mathrm{H}}_{\mathrm{M}^F_e}(x^{-2})$$

Case 1: e is not a coloop in M^F. We take the coefficient of $\emptyset$ in the formula (2).

$$x^f \underline{\mathrm{H}}_{\mathrm{M}^F}(x^{-2}) + x^{f-2}\underline{\mathrm{H}}_{\mathrm{M}^F_e}(x^{-2}) = x^f \underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(x^{-2}) + x^{-1} \sum_{G \in S_e(\mathrm{M}^F)} x^{f-g-1} \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(x^{-2}) \cdot x^g \underline{\mathrm{H}}_{\mathrm{M}^G}(x^{-2}).$$

Dividing by x^f and setting $y = x^{-2}$, we get:

$$\underline{\mathrm{H}}_{\mathrm{M}^F}(y) + y\underline{\mathrm{H}}_{\mathrm{M}^F_e}(y) = \underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(y) + y \sum_{G \in S_e(\mathrm{M}^F)} \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \underline{\mathrm{H}}_{\mathrm{M}^G}(y)$$

The sum on the right includes the term for $G = \emptyset$, which is $\underline{\mathrm{H}}_{\mathrm{M}^F_e}(y) \cdot \underline{\mathrm{H}}_{U_{0,0}}(y) = \underline{\mathrm{H}}_{\mathrm{M}^F_e}(y)$. Separating this term and canceling it with the $y\underline{\mathrm{H}}_{\mathrm{M}^F_e}(y)$ from the LHS yields the semi-small decomposition formula:

When e is not a coloop in M^F,

$$\underline{\mathrm{H}}_{\mathrm{M}^F}(y) = \underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(y) + \sum_{G \in S_e(\mathrm{M}^F)\setminus\{\emptyset\}} y\underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \underline{\mathrm{H}}_{\mathrm{M}^G}(y). \tag{4}$$

Case 2: e is a coloop in M^F. We take the coefficient of $\emptyset$ in the formula (3). Equating the coefficients gives:

$$x^f \underline{\mathrm{H}}_{\mathrm{M}^F}(x^{-2}) + x^{f-2}\underline{\mathrm{H}}_{\mathrm{M}^F_e}(x^{-2}) = (x + x^{-1})x^{f-1}\underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(x^{-2}) + x^{f-2}\sum_{\dots} \dots$$

Dividing by x^f and setting $y = x^{-2}$ leads to the known formula for adding a coloop:

When e is a coloop in M^F, (some errors here?? looks good to me)

$$\underline{\mathrm{H}}_{\mathrm{M}^F}(y) = (1+y)\underline{\mathrm{H}}_{\mathrm{M}^F\setminus e}(y) + \sum_{G \in \mathcal{L}(\mathrm{M}^F\setminus e)\setminus\{\emptyset\}} y\underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \underline{\mathrm{H}}_{\mathrm{M}^G}(y). \tag{5}$$

This completes the recovery of the known decomposition formulas of $\underline{\mathrm{H}}$.

3.1. Recovery of the known Augmented Chow Polynomial Decomposition Formulas

We consider another $\mathbb{Z}[x, x^{-1}]$-module homomorphism $\Phi_{\mathrm{M}} : \mathcal{H}(\mathrm{M}) \to \mathbb{Z}[x, x^{-1}]$ defined by

$$\Phi_{\mathrm{M}}(\alpha) = \sum_{F \in \mathcal{L}(\mathrm{M})} x^{-\mathrm{rk}(F)} \alpha_F,$$

which satisfies $\Phi_{\mathrm{M}\setminus e} \circ \Delta = \Phi_{\mathrm{M}}$. Applying this map to c^F gives

$$\Phi_{\mathrm{M}}(\mathrm{c}^F) = \sum_{G \leq F} x^{-\mathrm{rk}(G)} \mathrm{c}^F_G = x^{\mathrm{rk}(F)} \sum_{G \leq F} x^{-2\mathrm{rk}(G)} \underline{\mathrm{H}}_{\mathrm{M}^F_G}(x^{-2}) = x^{\mathrm{rk}(F)} \mathrm{H}_{\mathrm{M}^F}(x^{-2}) \in \mathrm{Pal}(0).$$

Remark. We see that $\Phi_{\mathrm{M}} = S^{\mathrm{M}}_{\emptyset}$. Thus, $S^{\mathrm{M}\setminus e}_{\emptyset} \circ \Delta = S^{\mathrm{M}}_{\emptyset}$.

Applying the $\Phi_{\mathrm{M}\setminus e}$ map to (2) yields the following identity:

$$x^f \mathrm{H}_{\mathrm{M}^F}(x^{-2}) = x^f \mathrm{H}_{\mathrm{M}^F\setminus e}(x^{-2}) + x^{-1} \sum_{G \in S_e(\mathrm{M}^F)} x^{f-g-1} \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(x^{-2}) \cdot x^g \mathrm{H}_{\mathrm{M}^G}(x^{-2}).$$

Simplifying the exponents and setting $y = x^{-2}$, we get:

When e is not a coloop in M^F,

$$\mathrm{H}_{\mathrm{M}^F}(y) = \mathrm{H}_{\mathrm{M}^F\setminus e}(y) + \sum_{G \in S_e(\mathrm{M}^F)} y \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \mathrm{H}_{\mathrm{M}^G}(y). \tag{6}$$

Proceeding similarly for the coloop case, we get the following identity:

$$x^f \mathrm{H}_{\mathrm{M}^F}(x^{-2}) = (x + x^{-1}) x^{f-1} \mathrm{H}_{\mathrm{M}^F\setminus e}(x^{-2}) + x^{-1} \sum_{G \in S_e(\mathrm{M}^F)} x^{f-g-1} \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(x^{-2}) \cdot x^g \mathrm{H}_{\mathrm{M}^G}(x^{-2}).$$

Simplifying the exponents and setting $y = x^{-2}$, we get:

When e is a coloop in M^F,

$$\mathrm{H}_{\mathrm{M}^F}(y) = (1+y) \mathrm{H}_{\mathrm{M}^F\setminus e}(y) + \sum_{G \in S_e(\mathrm{M}^F)} y \underline{\mathrm{H}}_{\mathrm{M}^F_{G\cup e}}(y) \cdot \mathrm{H}_{\mathrm{M}^G}(y). \tag{7}$$

4. References